

POLVIFIDEL: Evaluating and improving political grounding in VLM image captions

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Contributions

- 1. *Evaluation*: A political visual-fidelity metric** that evaluates whether captions capture politically meaningful visual content.
- 2. *Improvement*: A prompt-based caption improvement method** that uses domain-specific vision-model outputs as auxiliary information to produce politically grounded captions without fine-tuning.

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Working definitions

What is a politically grounded caption?



Caption A: A **man** in a suit is wearing a **cap**.

Caption B: **President Trump** is wearing a **MAGA hat**.



Caption A: President Trump **whispers** to President Xi.

Caption B: President Trump **yells** at President Xi.

A “politically-grounded” image caption should accurately capture political **actors**, symbolic **objects**, and the **relationships** connecting them.

What is a political object?

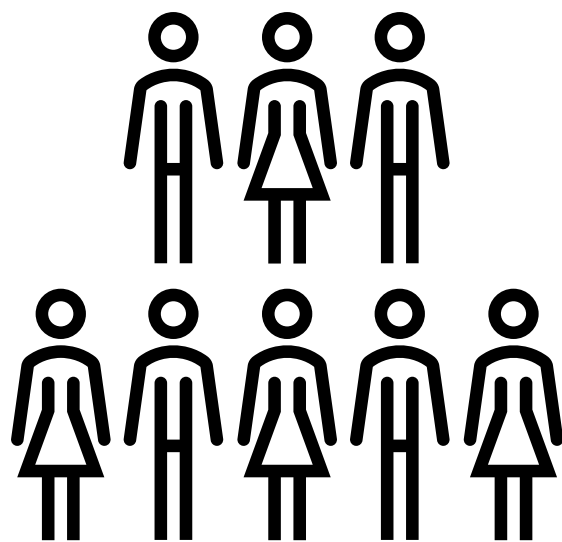


General Objects: {"person", "flag", "hat"}
Political Objects: {"civilian", "American flag," "MAGA hat", "Capitol Hill"}



General Objects: {"person", "flag"}
Political Objects: {"civilian", "Rainbow flag", "Supreme Court"}

Motivation & Problem Statement



Political Communication



Images



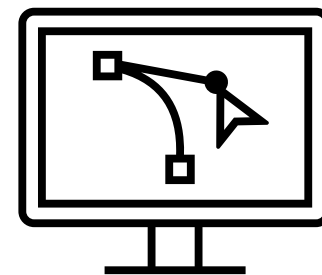
Political scientists

Q1: Are VLM-generated captions *politically grounded*?

Q2: If not, can we *improve* them?



Image descriptions
(captions)



VLM

Problem 1: Existing evaluation metrics cannot capture political grounding

Textual fidelity (Reference-based)

- Measures n-gram or semantic overlap between caption and reference **text**
 - e.g.) BLEU, METEOR, ROUGE, CIDEr, SPICE
- Fast
- Inherits human bias, omissions, and subjectivity

Visual fidelity (Reference-free)

- Measures overlap between caption and the input **image**
 - e.g.) VIFIDEL, CLIPScore
- Doesn't inherit human bias
- Slow
- Relies on general vision models



Reference: President Trump is wearing a MAGA hat.

Caption: A man is wearing a cap.

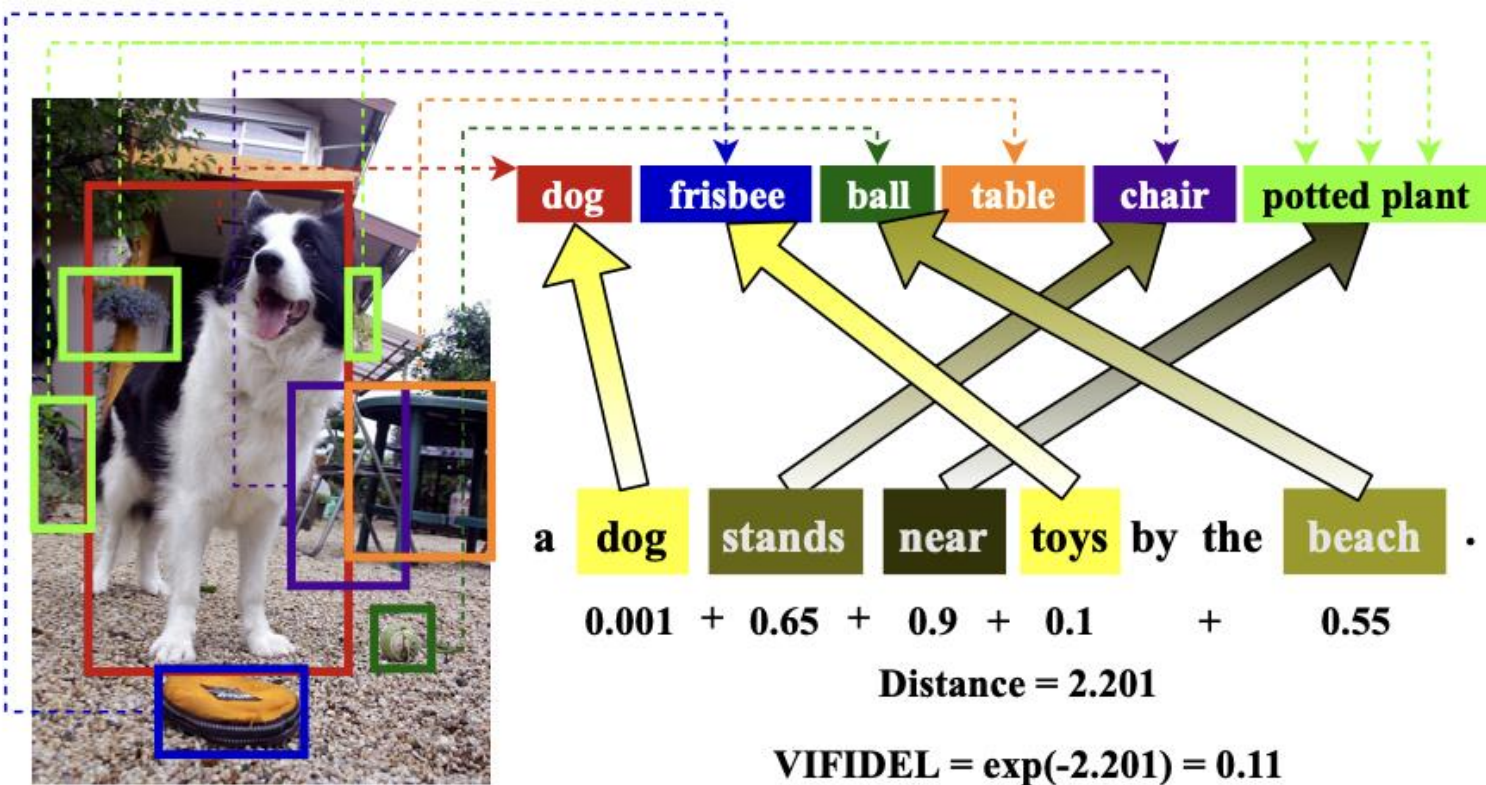
Problem 2: VLMs cannot reliably extract political actors and objects

Features of Interest	Related Vision Tasks	Can VLMs do it?	Are VLMs better than task-specific models?
Actors	Face Recognition		
Objects	Object Detection		
Relations	Scene Graph Generation Social Relationship Recognition Human Object Interaction		

Can we fuse precise detections from task-specific vision models with the flexible relational reasoning of VLMs to get the best of both worlds, without finetuning?

Part 1: Evaluation

A **political visual-fidelity metric** that evaluates whether captions capture politically meaningful visual content.



$$VIFIDEL(I, S) = \exp(-WMD(d^I, d^S))$$

$$WMD(d^I, d^S) = \min_{T \geq 0} \sum_{i,j=1}^N T_{ij} c(i, j)$$

$$c(i, j) = \|x_i - x_j\|_2^p$$

Strengths:

- **Reference-free** → avoids biases or errors in human captions
- **Fine-grained** → measures object-level fidelity

Limitations:

- Uses a **generic detector** → cannot detect political subcategories
- No weighting for **salience**
- No explicit **hallucination** penalty

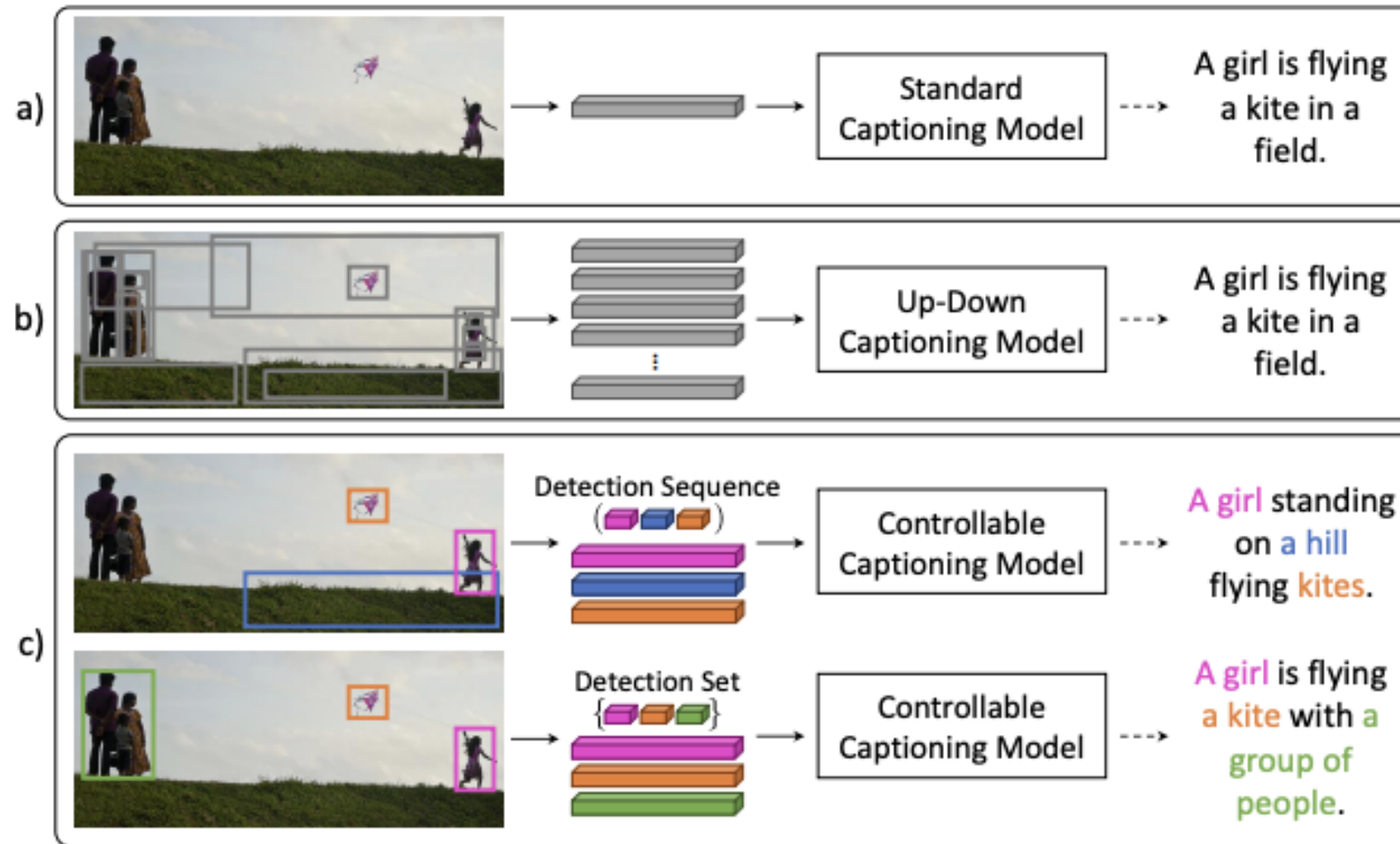
Madhyastha, Pranava Swaroop, Josiah Wang, and Lucia Specia. "VIFIDEL: Evaluating the visual fidelity of image descriptions." *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. 2019.

Proposed method 1: POLVIFIDEL

- POLVIFIDEL extends VIFIDEL in three ways:
 - Uses **domain-specific** face recognition and object detection models
 - Naturally handles entity **salience**
 - Explicitly incorporates **hallucination** penalty
- $POLVIFIDEL(I, S) = R_{obj} \cdot (1 - HAL)^\beta, \beta \geq 1$
 - $R_{obj} = \frac{\{Detected \cap Mentioned Political Entities\}}{\{Detected Political Entities\}}$
 - $HAL = \frac{\{Mentioned \cap Not Detected Political Entities\}}{\{Mentioned Political Entities\}}$

Part 2: Improvement

A **prompt-based caption improvement method** that uses vision-model outputs as auxiliary information to produce politically grounded captions without fine-tuning.



Strengths:

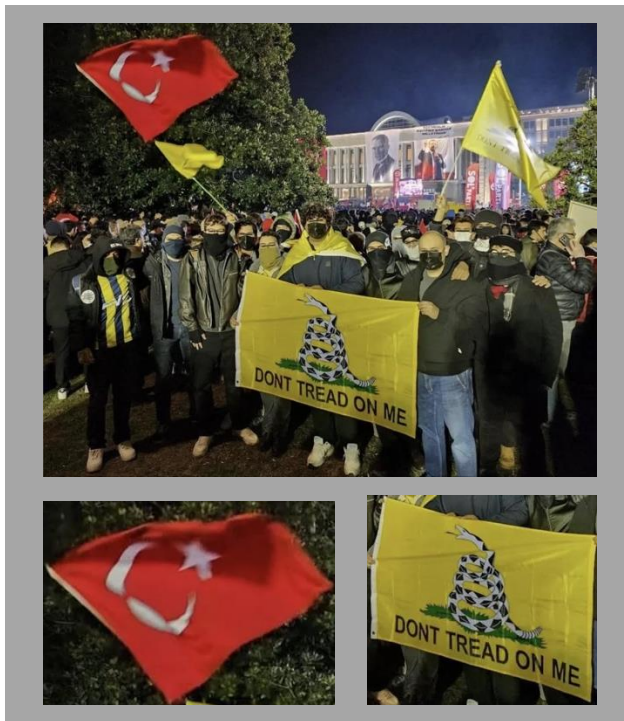
- Control captioning through region inputs

Limitations:

- Computationally expensive
- Not sure if VLMs understand the political meaning of provided regions

Proposed Method 2: Political objects as auxiliary information

Approach 1: Visual Aux



Approach 2: Textual Aux



Approach 3: Annotated Aux



Experiment plan &
Expected results

Experiment plan

- **Data:** 1200 NYT political images, stratified across political actors, ideological leanings, and event types.
- **Models:** 2 proprietary and 3 open-source VLMs
 - GPT-4o, Gemini 2.5, LLaVA-1.6, Intern-S1, Qwen3-VL
- **Procedure:**
 - Each model receives 4 prompts for each image
 - No Aux; Visual Aux; Textual Aux; Annotated Aux
 - Total: 24,000 captions
 - For each caption, compute:
 - 5 textual fidelity metrics: BLEU, CIDEr, ROUGE, METEOR, SPICE
 - 2 visual fidelity metric: VIFIDEL, POLVIFIDEL
 - Human visual-fidelity ratings: avg of three human ratings
 - Compute the correlation score between each metric and human judgments
 - See how downstream analysis changes with and without visual input

Expected outcomes

1. Evaluation

- POLVIFIDEL has the highest correlation with human judgements.

2. Caption Improvement

- Prompts with auxiliary object information produce higher human ratings and POLVIFIDEL scores.
- Biggest gains appear for low-salience politicians and events.

3. Downstream Effects

- Shifts topic modeling results

Thank you!